Gene Pulser® Electroprotocols

*** We recommend adapting this protocol to use the Gene Pulser electroporation buffer (catalog #165-2676, 165-2677), which increases cell viability and transfection efficiency in mammalian cell lines.**

Cell Type Mammalian

Species Used Human, K562, chronic myeloid leukemia

Molecules Electroporated DNA: pZ189

Before the Pulse

Cell Growth Medium RPMI 1640 (GIBCO/BRL,Sigma)

Growth Phase at Harvest 10 (6) cells / ml

Pre-pulse Incubation Not given

Wash Solution Phosphate Buffered Saline

The Pulse

Instruments Used Not given

Electroporation Temperature 20 to 25 °C (but pre-cooled on ice)

Electroporation Medium* Phosphate Buffered Saline

Cell Density 1 x 10 (7) cells / ml

Volume of Cells 0.4 ml

DNA Concentration 400 µg / ml

DNA Resuspension Buffer TE

Volume of DNA 40 µg

Cuvette Gap 0.4 cm

Voltage 0.6 kV

Field Strength 1.50 V/cm

Capacitor 25 µF

Resistor (Pulse Controller) Ω none

Time Constant 1.5 to 1.9 msec

After the Pulse

Outgrowth Medium RPMI 1640

Outgrowth Temperature 37 °C, 5% CO_2

Length of Incubation 1 day

Selection Method or Assay Used Not given

Electroporation Efficiency Not given

Per Cent Survival 50%

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.
PBS: 1x = 8g NaCl, 0.2g KCl, 0.2g KH_2PO_4, 1.15g Na_2HPO_4

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